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# BSc (Hons) in Information Technology Specializing Data Science

## Research Project - IT4010 Ethical Analysis Report

Group ID: 25-26J-127

Project Title: Legalvision: A Knowledge-Driven AI Framework for Legal Understanding & Trust Assessment

| | **Identification and Analysis** |
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| **Data Privacy Concerns** | The project uses real-world property law documents collected from legal professionals and publicly available legal sources. To protect privacy, all personally identifiable information (PII) such as names, addresses, and National Identity Card (NIC) numbers are replaced with dummy data during preprocessing. Any sensitive or confidential information is either anonymized or removed. This ensures that individuals' legal and personal privacy is preserved while still allowing the use of realistic legal data for analysis. |
| **Data Security** | To ensure data protection, all datasets are stored in secure environments with restricted access. Only authorized users can access the data during processing and model development. The system processes data in structured formats (CSV/Excel), and safeguards are applied to prevent unauthorized access or misuse. Additionally, outputs such as summaries and visualizations are generated without exposing raw legal documents directly. |
| **Bias and Fairness** | Since the dataset includes property law documents collected mainly from Sri Lankan legal sources, we check for any imbalances in document types and legal perspectives. The dataset is carefully prepared to include a variety of documents such as deeds, leases, and wills to ensure balanced representation. We also make adjustments during preprocessing and classification to handle variations in legal language and clause structures, ensuring fair and consistent interpretation across different legal contexts. |
| **Transparency and Accountability** | The system ensures transparency through a strong traceability mechanism, where each generated summary is directly linked to its original legal clauses. This allows users to verify outputs easily. Additionally, classification confidence scores are maintained, making model decisions interpretable rather than opaque. Clear documentation of data processing, model training, and evaluation is also maintained to ensure accountability. |
| **Impact on Stakeholders** | The system mainly benefits non-expert users, property owners, and general citizens by making complex legal documents easier to understand. It reduces dependency on legal professionals for basic interpretation and supports better decision-making. Legal professionals and businesses also benefit from improved efficiency. However, care must be taken to ensure users do not rely solely on automated summaries as legal advice, maintaining responsible use of the system. |